

UMC 0.11 um Logic and Mixed-Mode Metal Metal Capacitor Al Advanced Enhancement Process Resistor SPICE Model Document

**Version 0.6 Phase 1
2016/07/15**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
T.0 p1	2009/05/05	Kelly Pao	Original released.
T.1 p1	2009/05/20	Kelly Pao	1. Fixed ddw_rnpd_l110e form -1.6000e-008 to -8.4000e-008 of the ss, the drend_rnpd_l110e -1.9000e-5 to -2.2000e-005 of ff. 2. Added w*0.9 and l*0.9 in all resistor formulas.
T.2 p1	2009/12/11	Laney Fan	Salicide N+ diffusion ,Salicide P+ diffusion ,salicide N+ poly , salicide P+ poly and NWell resistors are copied from L110E_RES_V031
0.1 p1	2010/01/08	Jian Zhi	1. Modify vcr1 and vcr2 parameters in the Non-salicide N+Poly, Non-salicide P+Poly, Non-salicide N+Diffusion, Non-salicide P+Diffusion and Non-salicide HR Poly resistor devices. 2. Modify didoe parameters in the Non-salicide N+Diffusion, Non-salicide P+Diffusion, Salicide N+ diffusion, Salicide P+ diffusion and NWell resistors. 3. New DSM version definition from 2009/11/30.
0.2 p1	2010/07/07	Jian Zhi	Modify rend and dw in the Non-salicide N+ Poly, Non-salicide P+ Poly, Non-salicide N+ Diffusion, Non-salicide P+ Diffusion and Non-salicide HR Poly resistor devices.
0.3 p1	2010/11/03	Shih-Hsin Yeh	1. Add M1~M6 resistor models. 2. Revise w and l to (w*0.9) and (l*0.9) in salicide N+/P+ poly resistor models.
0.3 p2	2011/1/24	Shih-Hsin Yeh	Modify the dimension range to align with TLR.
0.4 p1	2014/11/18	Bing Chang Wu	Revise BEOL metal resistor models as r_m1, r_1x, r_2x
0.5 p1	2015/1/26	Bing Chang Wu	Update r_2x Rs from 0.04 to 0.035 ohm/sq
0.6 p1	2016/07/15	Tzu Kang	1. Add flicker noise model of

		Hsiung	rnnpo_l110e , rnppo_l110e , rnhr1000_l110e. 2. Revise diode parameters according to l110ae_diode_v061 . 3. Revise voltage coefficient of rnnd_l110e , rnpd_l110e according to the original measurement data. 4. Add Model equation and Parameter into the document .
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1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode Metal Metal Capacitor Al Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2014.9-SP2
Cadence SPECTRE version 15.1.0.257 64bit
Mentor Graphics ELDO version 15.4

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

tt: Typical resistor case
ss: Maximum resistor case
ff: Minimum resistor case

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator.
.option scale=0.9
.lib "/ModelFileName.lib" CaseName
, where CaseName could be tt, ss, or ff.
- Define device condition by 'Xxx' statement.
e.g. Xres Port1 Port2 Well rnnpo_l110e l=2u w=5u

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator.
SetOption1 options scale=0.9
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName could be tt, ss, or ff.
- Define device condition by 'Xxx' statement.
e.g. Xres (Port1 Port2 Well) rnnpo_l110e l=2u w=5u

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator.
.option scale=0.9
.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName could be tt, ss, or ff.
- Define device condition by 'Xxx' statement.
e.g. Xres Port1 Port2 Well rnnpo_l110e l=2u w=5u

ModelFileName: l110ae_res_v061

The available resistors are:

- a. Non-salicide N+ Poly resistor : rnnpo_l110e
- b. Non-salicide P+ Poly resistor : rnppo_l110e
- c. Non-salicide N+ Diffusion resistor : rnnd_l110e
- d. Non-salicide P+ Diffusion resistor : rnpd_l110e
- e. Non-salicide HR Poly resistor : rnhr1000_l110e
- f. Salicide N+ diffusion resistor : rsnd_mml110e
- g. Salicide P+ diffusion resistor : rspd_mml110e
- h. Salicide N+ poly resistor : rsnp_mml110e
- i. Salicide P+ poly resistor : rspp_mml110e
- j. NWell resistor : rnwell_mml110e
- k. M1 resistor : r_m1_l110e
- l. 1x resistor : r_1x_l110e
- m. 2x with 8k metal resistor : r_2x_l110e

Geometry range:(before 90% shrinkage)

- a. Non-salicide N+ Poly resistor : rnnpo_l110e
 $1\ \mu\text{m} < \text{LDES} < 18\ \mu\text{m}$
 $0.5\ \mu\text{m} < \text{WDES} \leq 20\ \mu\text{m}$
- b. Non-salicide P+ Poly resistor : rnppo_l110e

$$1 \mu\text{m} < \text{LDES} < 18 \mu\text{m}$$
$$0.5 \mu\text{m} < \text{WDES} \leq 20 \mu\text{m}$$

c. Non-salicide N+ Diffusion resistor : rnnd_l110e

$$1 \mu\text{m} < \text{LDES} < 18 \mu\text{m}$$
$$0.5 \mu\text{m} < \text{WDES} \leq 20 \mu\text{m}$$

d. Non-salicide P+ Diffusion resistor : rnpd_l110e

$$1 \mu\text{m} < \text{LDES} < 18 \mu\text{m}$$
$$0.5 \mu\text{m} < \text{WDES} \leq 20 \mu\text{m}$$

e. Non-salicide HR Poly resistor : rnhr1000_l110e

$$1 \mu\text{m} < \text{LDES} < 18 \mu\text{m}$$
$$0.5 \mu\text{m} < \text{WDES} \leq 20 \mu\text{m}$$

f. NWell resistor : rnwell_mml110e

$$1.67 \mu\text{m} < \text{WDES} \leq 20 \mu\text{m}$$

Bias Range:

a. Non-salicide

Jmax <= 1.0mA/um for non-salicide diffusion resistor.

Jmax <= 0.5mA/um for non-salicide poly resistor.

b. Salicide

Jmax <= 2.0mA/um for salicide diffusion resistor.

Jmax <= 2.0mA/um for salicide poly resistor.

c. Nwell

Jmax <= 0.05mA/um for Nwell resistor.

Jmax: Maximum current density of the resistor.

Note: Applied voltage max

$$V = J_{\text{max}} * \text{Width} * \text{Resistance}$$

Temperature range:

-55 °C ~ +135 °C

Equation to Estimate Maximum Voltage:

The following equation can be used to estimate the maximum Voltage for given dimension (l, w)

For non-salicide N+/P+ poly resistor

$$V_{max} = J_{max} * w * 1e6 * ((2 * r_{end} / (w * 0.9 + dw)) + (r_s * l * 0.9 / (w * 0.9 + dw)))$$

For non-salicide N+/P+ diffusion resistor

$$V_{max} = J_{max} * w * 1e6 * ((2 * r_{end} / (w * 0.9 + dw)) + (r_s * l * 0.9 / (w * 0.9 + dw)))$$

For non-salicide HR Poly resistor

$$V_{max} = J_{max} * w * 1e6 * ((2 * r_{end} / (w * 0.9 + dw)) + (r_s * ((l * 0.9) - 0.8e-6) / (w * 0.9 + dw)))$$

For salicide N+/P+ diffusion resistor

$$V_{max} = J_{max} * w * 1e6 * (r_s * l * 0.9 / (w * 0.9 + dw))$$

For salicide N+/P+ poly resistor

$$V_{max} = J_{max} * w * 1e6 * (r_s * l * 0.9 / (w * 0.9 + dw))$$

For N-Well Resistor :

$$V_{max} = J_{max} * w * 1e6 * (r_s * l * 0.9 / (w * 0.9 + dw))$$

Device name	jmax(A/μm)	rs(ohm/sq)	rend(ohm*m)	dw(m)
rnpno_l110e	5e-4	113	1.50e-5	-7.05e-8
rnppo_l110e	5e-4	239	3.650e-5	-4.05e-8
rnnd_l110e	1e-3	95	1.34e-5	-1.70e-8
rnpd_l110e	1e-3	115	1.07e-05	-1.50e-8
rnhr1000_l110e	5e-4	984	2.02e-4	-4.70e-8
rsnd_mml110e	2e-3	8.30	--	0.00
rspd_mml110e	2e-3	7.20	--	0.00
rsnp_mml110e	2e-3	8.70	--	0.00
rspp_mml110e	2e-3	7.30	--	0.00
rnwell_mml110e	5e-5	370.0	--	-5.10e-07

2 Model equation and Parameter

2.1 Non-Salicide N+ Poly Resistor

$$R = R_0 * [1 + VCR1 * \frac{V}{L} + VCR2 * \left(\frac{V}{L}\right)^2] * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{2 * R_{end} + R_s * L}{W + \Delta W}$$

where:

R: total resistance in ohm
R₀: intrinsic resistance at 25 °C in ohm
R_s: sheet resistance at 25 °C in ohm/sq
R_{end}: end resistance at 25 °C in ohm*m
W: designed non-salicide resistor width in μm
L: designed non-salicide resistor SAB length in μm
Δ*W*: deviation of designed resistor width bias at 25 °C in m
V: bias voltage across resistor
VCR1: linear voltage coefficient of resistor in m/V for N+ poly resistor
VCR2: quadratic voltage coefficient of resistor in m²/V² for N+ poly resistor
T: device temperature in °C (T₀ = 25°C)
TC1: linear temperature coefficient of resistor in 1/ °C
TC2: quadratic temperature coefficient of resistor in 1/ °C²

	FF.	TT.	SS.	Unit
N+ Poly <i>R_s</i>	85	113	141	Ohm/sq
N+ Poly <i>R_{end}</i>	3.010E-6	1.50E-5	2.702E-5	Ohm*m
N+ Poly Δ <i>W</i>	-5.45E-8	-7.05E-8	-8.55E-8	m
N+ Poly <i>VCR1</i>	--	0.00	--	m/V
N+ Poly <i>VCR2</i>	--	-2.58E-14	--	m ² /V ²
N+ Poly <i>TC1</i>	--	1.60E-04	--	1/ °C
N+ Poly <i>TC2</i>	--	2.81E-07	--	1/ °C ²

2.2 Non-Salicide P+ Poly Resistor

$$R = R_0 * [1 + VCR1 * \frac{V}{L} + VCR2 * \left(\frac{V}{L}\right)^2] * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{2 * R_{end} + R_s * L}{W + \Delta W}$$

where:

R: total resistance in ohm

*R*₀: intrinsic resistance at 25 °C in ohm

*R*_s: sheet resistance at 25 °C in ohm/sq

*R*_{end}: end resistance at 25 °C in ohm*m

W: designed non-salicide resistor width in μm

L: designed non-salicide resistor SAB length in μm

Δ*W*: deviation of designed resistor width bias at 25 °C in m

V: bias voltage across resistor

VCR1: linear voltage coefficient of resistor in m/V for P+ poly resistor

VCR2: quadratic voltage coefficient of resistor in m²/V² for P+ poly resistor

T: device temperature in °C (T₀ = 25°C)

TC1: linear temperature coefficient of resistor in 1/ °C

TC2: quadratic temperature coefficient of resistor in 1/ °C²

	FF.	TT.	SS.	Unit
P+ Poly <i>R</i> _s	179	239	299	Ohm/sq
P+ Poly <i>R</i> _{end}	2.047E-5	3.65E-5	5.197E-5	Ohm*m
P+ Poly Δ <i>W</i>	-2.45E-8	-4.05E-8	-5.55E-8	m
P+ Poly <i>VCR1</i>	--	0.00	--	m/V
P+ Poly <i>VCR2</i>	--	5.86E-14	--	m ² /V ²
P+ Poly <i>TC1</i>	--	9.67E-05	--	1/ °C
P+ Poly <i>TC2</i>	--	5.60E-07	--	1/ °C ²

2.3 Non-Salicide N+ Diffusion Resistor

$$R = R_0 * [1 + VCR1 * V + VCR2 * (V)^2] * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{2 * R_{end} + R_s * L}{W + \Delta W}$$

where:

R : total resistance in ohm

R_0 : intrinsic resistance at 25 °C in ohm

R_s : sheet resistance at 25 °C in ohm/sq

R_{end} : end resistance at 25 °C in ohm*m

W : designed non-salicide resistor width in μm

L : designed non-salicide resistor SAB length in μm

ΔW : deviation of designed resistor width bias at 25 °C in m

V : bias voltage across resistor

$VCR1$: linear voltage coefficient of resistor in 1/V for N+ diffusion resistor

$VCR2$: quadratic voltage coefficient of resistor in 1/V² for N+ diffusion resistor

T : device temperature in °C ($T_0 = 25^\circ\text{C}$)

$TC1$: linear temperature coefficient of resistor in 1/ °C

$TC2$: quadratic temperature coefficient of resistor in 1/ °C²

	FF.	TT.	SS.	Unit
N+ Diffusion R_s	71.0	95.0	119.0	Ohm/sq
N+ Diffusion R_{end}	7.810E-6	1.34E-5	1.897E-5	Ohm*m
N+ Diffusion ΔW	3.40E-8	-1.70E-8	-6.80E-8	m
N+ Diffusion $VCR1$	--	0.00	--	1/V
N+ Diffusion $VCR2$	--	1.46E-02	--	1/V ²
N+ Diffusion $TC1$	--	1.41E-03	--	1/ °C
N+ Diffusion $TC2$	--	4.35E-07	--	1/ °C ²

2.4 Non-Salicide P+ Diffusion Resistor

$$R = R_0 * [1 + VCR1 * V + VCR2 * (V)^2] * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{2 * R_{end} + R_s * L}{W + \Delta W}$$

where:

R : total resistance in ohm

R_0 : intrinsic resistance at 25 °C in ohm

R_s : sheet resistance at 25 °C in ohm/sq

R_{end} : end resistance at 25 °C in ohm*m

W : designed non-salicide resistor width in μm

L : designed non-salicide resistor SAB length in μm

ΔW : deviation of designed resistor width bias at 25 °C in m

V : bias voltage across resistor

$VCR1$: linear voltage coefficient of resistor in 1/V for P+ diffusion resistor

$VCR2$: quadratic voltage coefficient of resistor in 1/V² for P+ diffusion resistor

T : device temperature in °C ($T_0 = 25^\circ\text{C}$)

$TC1$: linear temperature coefficient of resistor in 1/ °C

$TC2$: quadratic temperature coefficient of resistor in 1/ °C²

	FF.	TT.	SS.	Unit
P+ Diffusion R_s	86.0	115.0	144.0	Ohm/sq
P+ Diffusion R_{end}	3.830E-6	1.07E-5	1.665E-5	Ohm*m
P+ Diffusion ΔW	3.60E-8	-1.50E-8	-6.60E-8	m
P+ Diffusion $VCR1$	--	0.00	--	1/V
P+ Diffusion $VCR2$	--	1.43E-02	--	1/V ²
P+ Diffusion $TC1$	--	1.47E-03	--	1/ °C
P+ Diffusion $TC2$	--	6.31E-07	--	1/ °C ²

2.5 HR Poly Resistor

$$R = R_0 * [1 + VCR1 * \frac{V}{L} + VCR2 * (\frac{V}{L})^2] * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{2 * R_{end} + Rs * (L - 0.8E - 06)}{W + \Delta W}$$

where:

R: total resistance in ohm
R₀: intrinsic resistance at 25 °C in ohm
Rs: sheet resistance at 25 °C in ohm/sq
R_{end}: end resistance at 25 °C in ohm*m
W: designed HR poly resistor width in m
L: designed HR poly resistor length in m
 ΔW : deviation of designed resistor width bias at 25 °C in m
V: bias voltage across resistor
VCR1: linear voltage coefficient of resistor in m/V for HR poly resistor
VCR2: quadratic voltage coefficient of resistor in m²/V² for HR poly resistor
T: device temperature in °C (T₀ = 25°C)
TC1: linear temperature coefficient of resistor in 1/ °C
TC2: quadratic temperature coefficient of resistor in 1/ °C²

	FF	TT	SS	Unit
HR poly Rs	836.0	984.00	1132.0	Ohm/sq
HR poly Rend	1.4869E-4	2.02E-4	2.5674E-4	Ohm*m
HR poly ΔW	-3.40E-8	-4.70E-8	-6.0E-8	m
HR poly VCR1	--	0.00	--	m/V
HR poly VCR2	--	-1.07E-13	--	m ² /V ²
HR poly TC1	--	-4.96E-04	--	1/ °C
HR poly TC2	--	1.25E-06	--	1/ °C ²

2.6 Salicide N+ Diffusion Resistor

$$R = R_0 * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{Rs * L}{W + \Delta W}$$

where:

R : total resistance in ohm

R_0 : intrinsic resistance at 25 °C in ohm

Rs : sheet resistance at 25 °C in ohm/sq

W : designed salicide resistor width in μm

L : designed salicide resistor length in μm

ΔW : deviation of designed resistor width bias at 25 °C in m

V : bias voltage across resistor

T : device temperature in °C ($T_0 = 25^\circ C$)

$TC1$: linear temperature coefficient of resistor in 1/ °C

$TC2$: quadratic temperature coefficient of resistor in 1/ °C²

	FF.	TT.	SS.	Unit
N+ Diffusion Rs	2.0	8.30	14.0	Ohm/sq
N+ Diffusion ΔW		0.0		m
N+ Diffusion TC1	--	2.879E-03	--	1/ °C
N+ Diffusion TC2	--	1.379E-06	--	1/ °C ²

2.7 Salicide P+ Diffusion Resistor

$$R = R_0 * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{Rs * L}{W + \Delta W}$$

where:

- R*: total resistance in ohm
- R₀*: intrinsic resistance at 25 °C in ohm
- Rs*: sheet resistance at 25 °C in ohm/sq
- W*: designed salicide resistor width in μm
- L*: designed salicide resistor length in μm
- ΔW*: deviation of designed resistor width bias at 25 °C in m
- V*: bias voltage across resistor
- T*: device temperature in °C (T₀ = 25°C)
- TC1*: linear temperature coefficient of resistor in 1/ °C
- TC2*: quadratic temperature coefficient of resistor in 1/ °C²

	FF.	TT.	SS.	Unit
P+ Diffusion Rs	2.0	7.20	14.0	Ohm/sq
P+ Diffusion ΔW		0.0		m
P+ Diffusion TC1	--	2.891E-03	--	1/ °C
P+ Diffusion TC2	--	1.457E-06	--	1/ °C ²

2.8 Salicide N+ Poly Resistor

$$R = R_0 * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{Rs * L}{W + \Delta W}$$

where:

R : total resistance in ohm

R_0 : intrinsic resistance at 25 °C in ohm

Rs : sheet resistance at 25 °C in ohm/sq

W : designed salicide resistor width in μm

L : designed salicide resistor length in μm

ΔW : deviation of designed resistor width bias at 25 °C in m

V : bias voltage across resistor

T : device temperature in °C ($T_0 = 25^\circ\text{C}$)

$TC1$: linear temperature coefficient of resistor in $1/^\circ\text{C}$

$TC2$: quadratic temperature coefficient of resistor in $1/^\circ\text{C}^2$

	FF.	TT.	SS.	Unit
N+ Poly Rs	2.00	8.70	15.0	Ohm/sq
N+ Poly ΔW		0.00		m
N+ Poly TC1	--	2.667E-03	--	$1/^\circ\text{C}$
N+ Poly TC2	--	1.163E-06	--	$1/^\circ\text{C}^2$

2.9 Salicide P+ Poly Resistor

$$R = R_0 * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{Rs * L}{W + \Delta W}$$

where:

R : total resistance in ohm

R_0 : intrinsic resistance at 25 °C in ohm

Rs : sheet resistance at 25 °C in ohm/sq

W : designed salicide resistor width in μm

L : designed salicide resistor length in μm

ΔW : deviation of designed resistor width bias at 25 °C in m

V : bias voltage across resistor

T : device temperature in °C ($T_0 = 25^\circ\text{C}$)

$TC1$: linear temperature coefficient of resistor in $1/^\circ\text{C}$

$TC2$: quadratic temperature coefficient of resistor in $1/^\circ\text{C}^2$

	FF.	TT.	SS.	Unit
P+ Poly Rs	2.0	7.30	15.0	Ohm/sq
P+ Poly ΔW		0.00		m
P+ Poly TC1	--	2.746E-03	--	$1/^\circ\text{C}$
P+ Poly TC2	--	1.631E-06		$1/^\circ\text{C}^2$

2.10 N-WELL Resistor

$$R = R_0 * [1 + (VC0 + VC1 * \min(V) + VC2 * (\min(V))^2) * |V|] * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{Rs * L}{W + \Delta W}$$

where:

R : total resistance in ohm

R_0 : intrinsic resistance at 25 °C in ohm

Rs : sheet resistance at 25 °C in ohm/sq

W : designed N-well resistor width in μm

L : designed N-well resistor length in μm

ΔW : deviation of designed resistor width bias at 25 °C in m

V : bias voltage across resistor

$VC0$: zero order voltage coefficient of resistor in 1/V for N-WELL resistor

$VC1$: first order voltage coefficient of resistor in 1/V² for N-WELL resistor

$VC2$: second order voltage coefficient of resistor in 1/V³ for N-WELL resistor

T : device temperature in °C ($T_0 = 25^\circ\text{C}$)

$TC1$: linear temperature coefficient of resistor in 1/ °C

$TC2$: quadratic temperature coefficient of resistor in 1/ °C²

	FF.	TT.	SS.	Unit
N-well Rs	321.6	370.0	419.0	Ohm/sq
N-well ΔW	--	-5.10E-07	--	m
N-well VC0	--	1.9571E-03	--	1/V
N-well VC1	--	-3.4571E-04	--	1/V ²
N-well VC2	--	5.7143E-05	--	1/V ³
N-well TC1	--	1.485E-03	--	1/ °C
N-well TC2	--	9.097E-06	--	1/ °C ²

2.11 Metal Resistor

$$R = R_0 * [1 + TC1 * (T - 25) + TC2 * (T - 25)^2]$$

$$R_0 = \frac{Rs * L}{W}$$

where:

R : total resistance in ohm
 R_0 : intrinsic resistance at 25 °C in ohm
 Rs : sheet resistance at 25 °C in ohm/sq
 W : designed metal resistor width in μm
 L : designed metal resistor length in μm
 ΔW : deviation of designed resistor width bias at 25 °C in m
 T : device temperature in °C ($T_0 = 25^\circ C$)
 $TC1$: linear temperature coefficient of resistor in 1/ °C
 $TC2$: quadratic temperature coefficient of resistor in 1/ °C²

	Min.	Typ.	Max.	Unit
M1 resistor Rs(W=0.144um)	105	170	225	mOhm/sq
1X Metal Rs(W=0.18um)	85	130	175	mOhm/sq
2X with 8k Metal Rs(W=0.396um)	5	35	75	mOhm/sq
M1 resistor TC1 (W=0.144um)	--	3.203E-03	--	1/ °C
1X Metal TC1 (W=0.18um)	--	3.304E-03	--	1/ °C
2X with 8k Metal TC1(W=0.396um)	--	3.415E-03	--	1/ °C
M1 resistor TC2 (W=0.144um)	--	-4.119E-07	--	1/ °C ²
1X Metal TC2 (W=0.18um)	--	-4.938E-07	--	1/ °C ²
2X with 8k Metal TC2(W=0.396um)	--	-2.26E-07	--	1/ °C ²

3.1.2 P+ Poly Resistor WAT Trend Chart

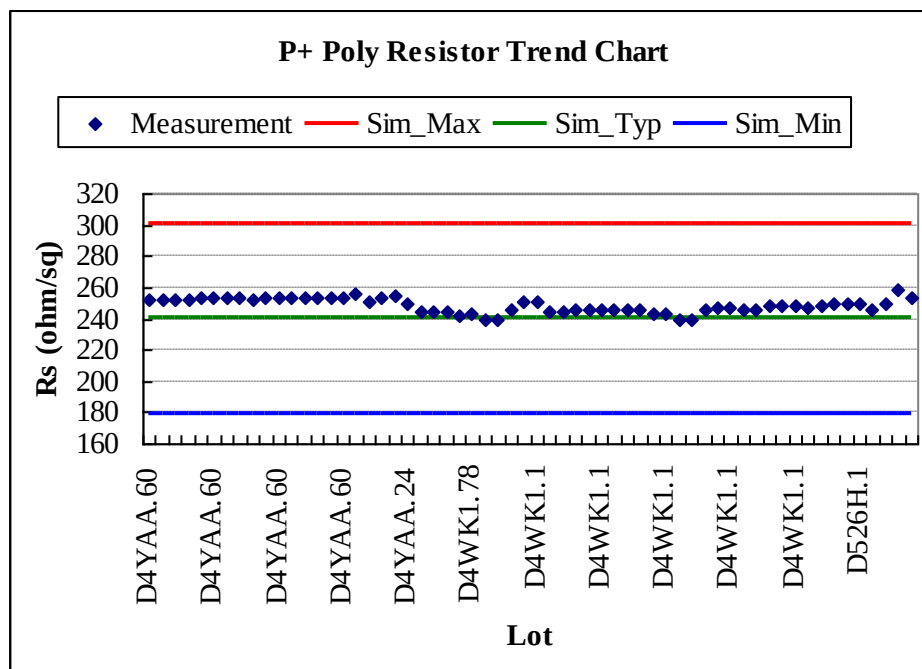


Figure 3-2 P+ Poly Resistor WAT Trend Chart

3.1.3 N+ Diffusion Resistor WAT Trend Chart

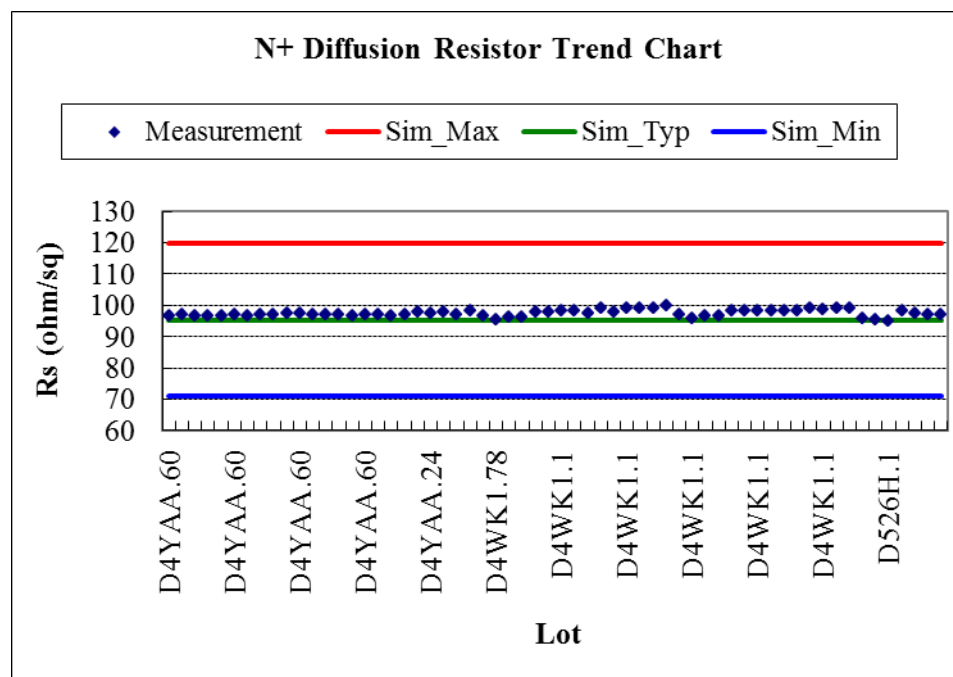


Figure 3-3 N+ Diffusion Resistor WAT Trend Chart

3.1.4 P+ Diffusion Resistor WAT Trend Chart

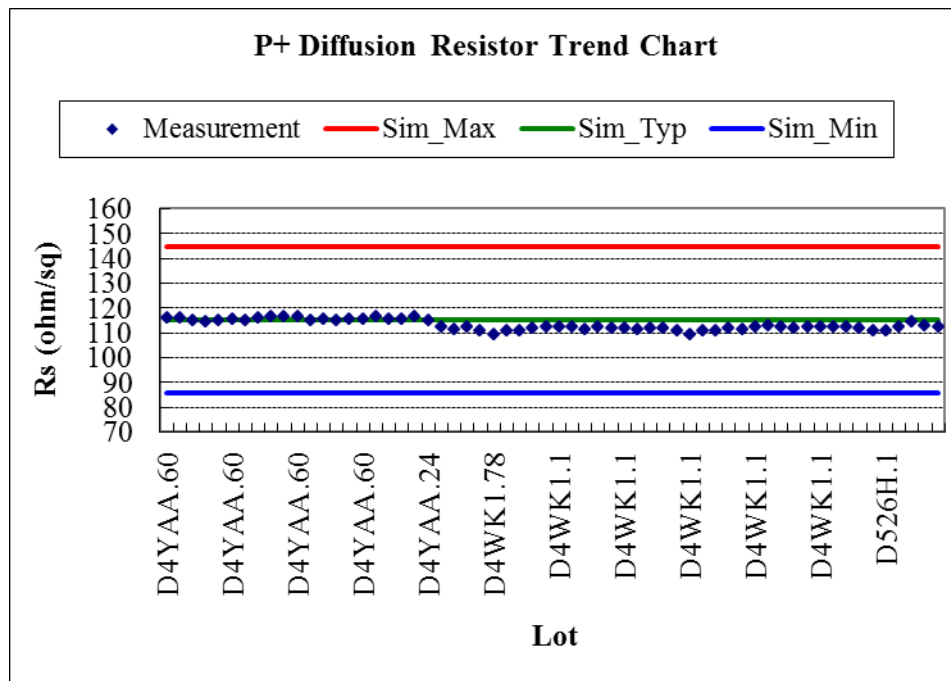


Figure 3-4 P+ Diffusion Resistor WAT Trend Chart

3.1.5 HR Resistor WAT Trend Chart

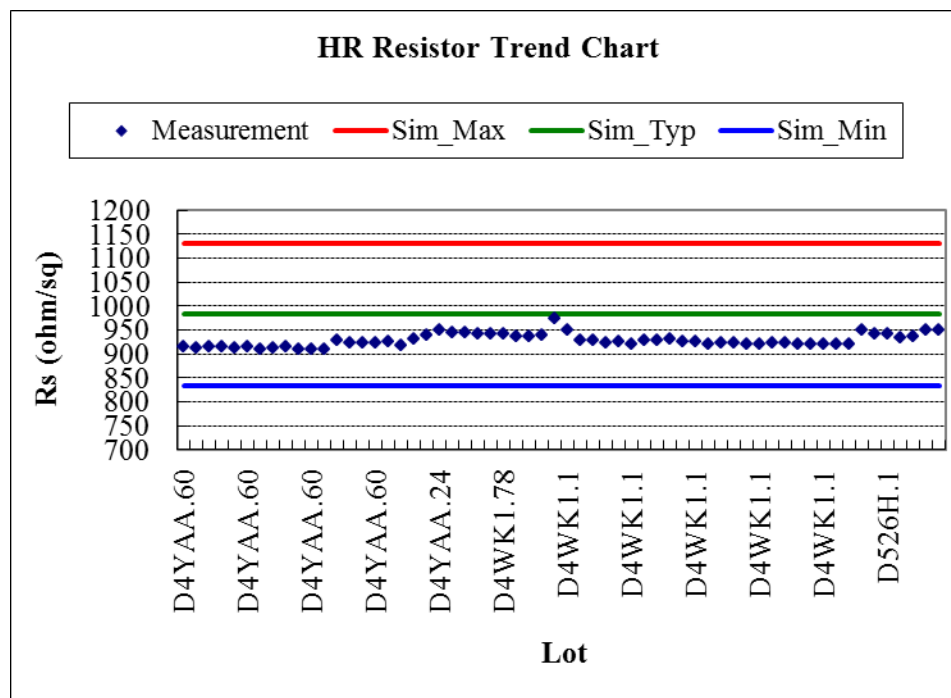


Figure 3-5 HR Resistor WAT Trend Chart

3.1.6 M1 Resistor WAT Trend Chart

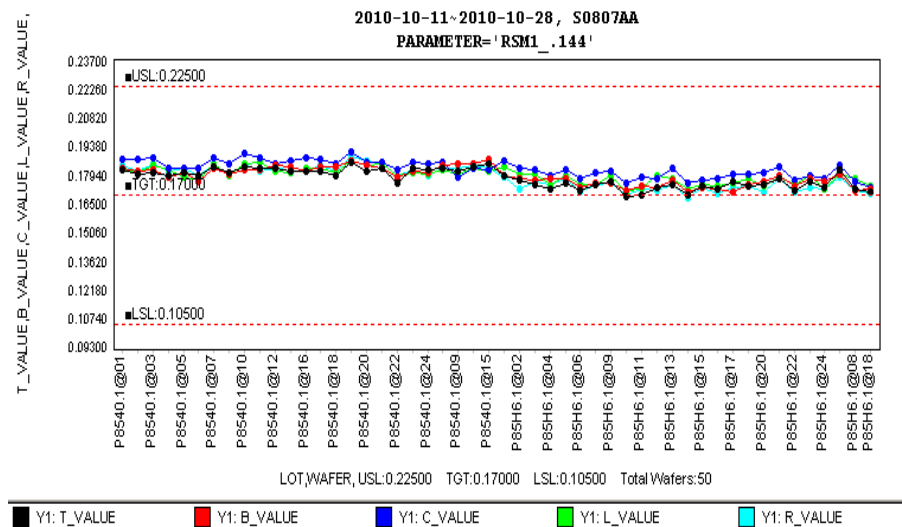


Figure 3-6 M1 Resistor WAT Trend Chart

3.1.7 M2 Resistor WAT Trend Chart

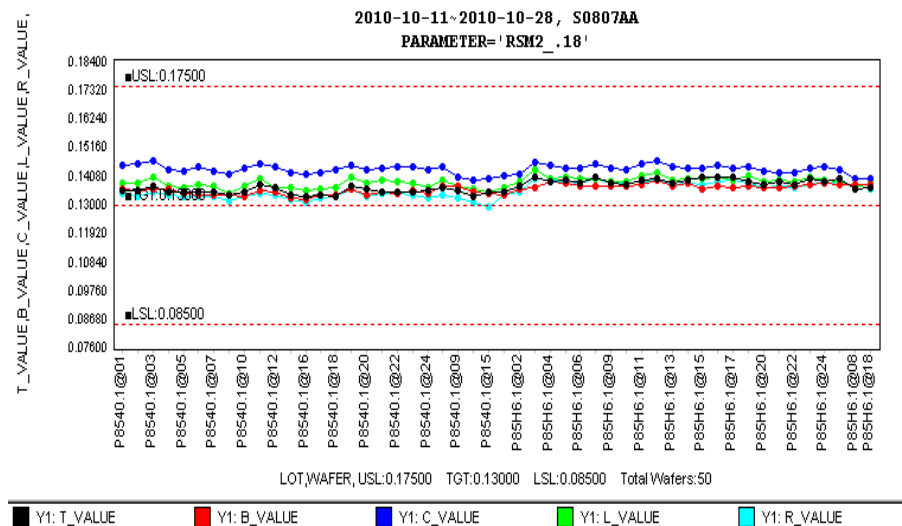


Figure 3-7 M2 Resistor WAT Trend Chart

3.1.8 M3 Resistor WAT Trend Chart

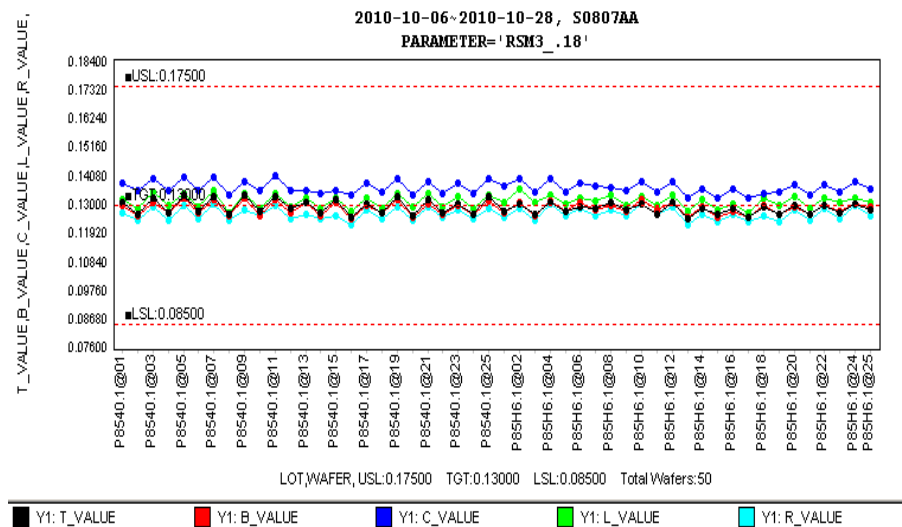


Figure 3-8 M3 Resistor WAT Trend Chart

3.1.9 M4 Resistor WAT Trend Chart

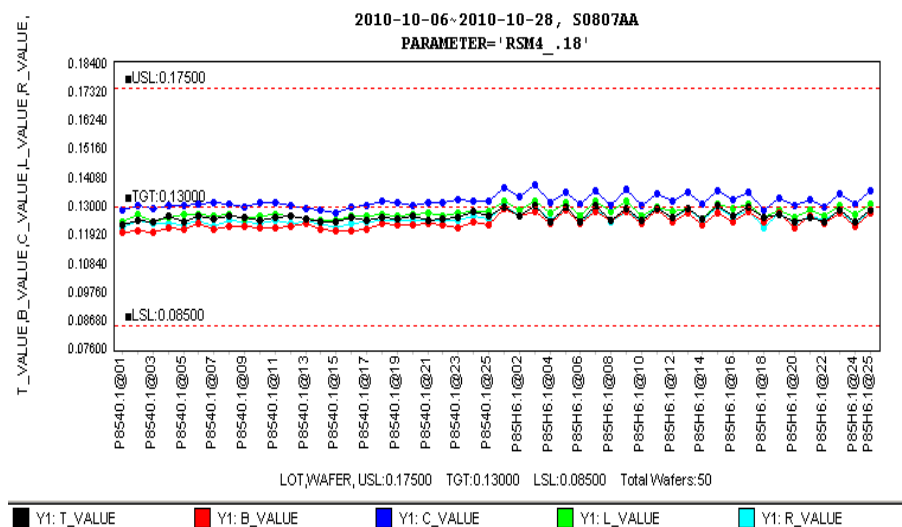


Figure 3-9 M4 Resistor WAT Trend Chart

3.1.10 M5 Resistor WAT Trend Chart

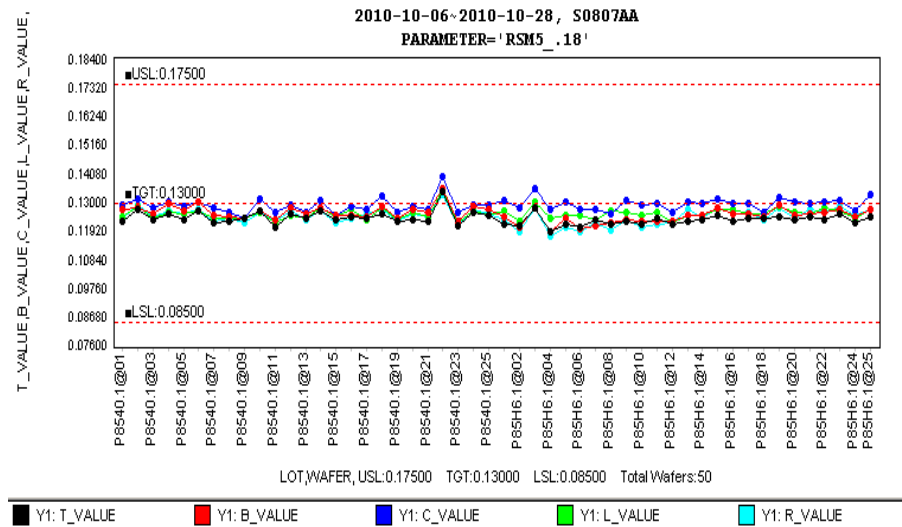


Figure 3-10 M5 Resistor WAT Trend Chart

4 Noise Characteristic

4.1 Verification plots

4.1.1 Non Salicide N+ Poly Resistor

➤ $W=0.45\mu\text{m}$ $L=90\mu\text{m}$ (After shrink)

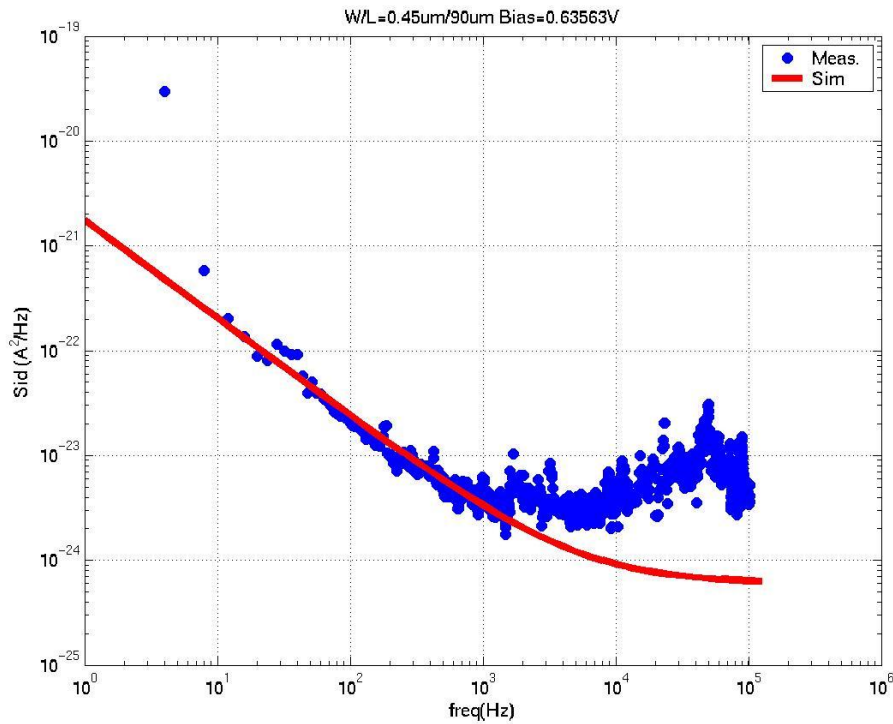


Figure 4-1 Non-Salicide N+ Poly Resistor W/L=0.45 μm /90 μm @ V=0.6356V

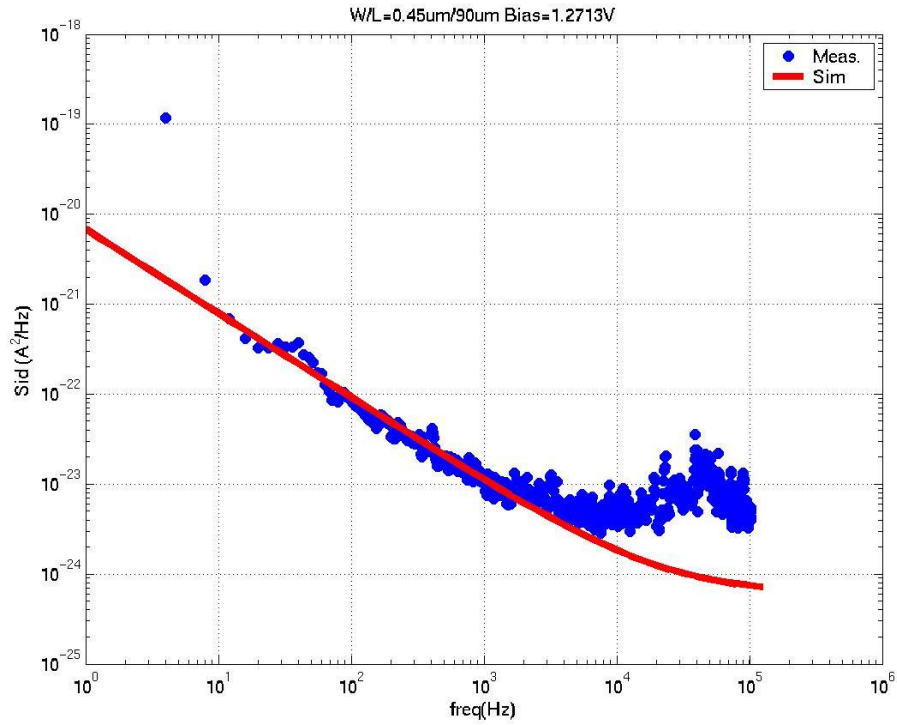


Figure 4-2 Non-Salicide N+ Poly Resistor W/L=0.45μm/90μm @ V=1.2713V

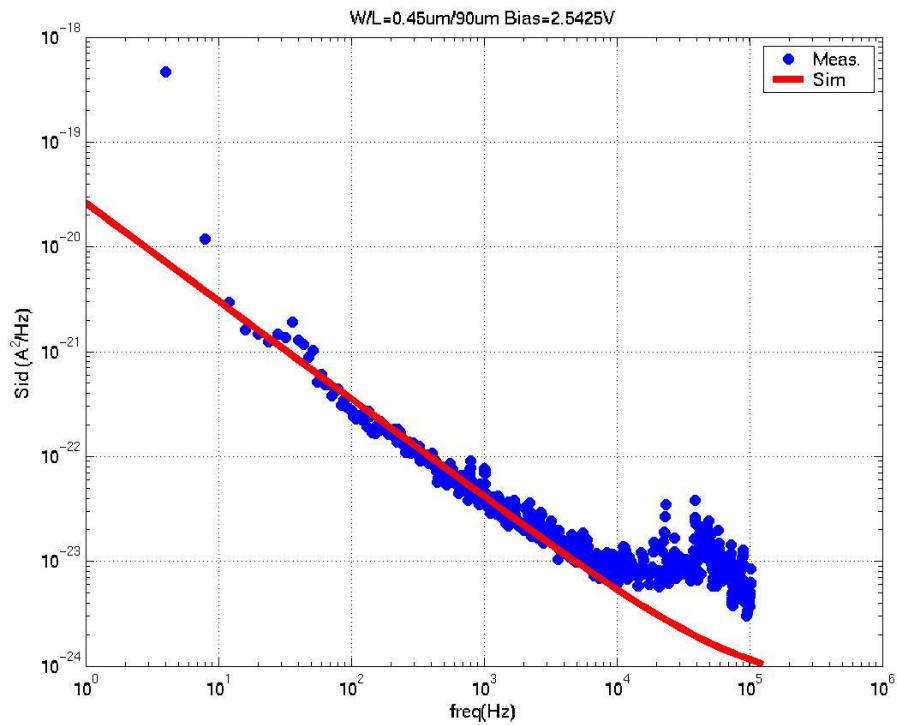


Figure 4-3 Non-Salicide N+ Poly Resistor W/L=0.45μm/90μm @ V=2.5425V

4.1.2 Non Salicide P+ Poly Resistor

➤ $W=0.45\mu\text{m}$ $L=90\mu\text{m}$ (After shrink)

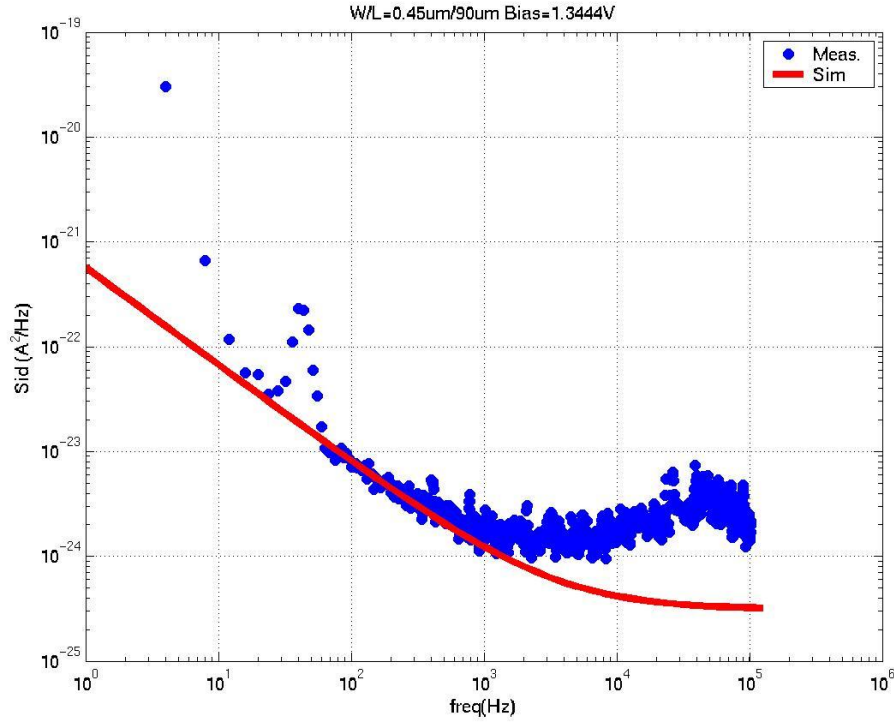


Figure 4-4 Non-Salicide P+ Poly Resistor W/L=0.45 μm /90 μm @ V=1.3444V

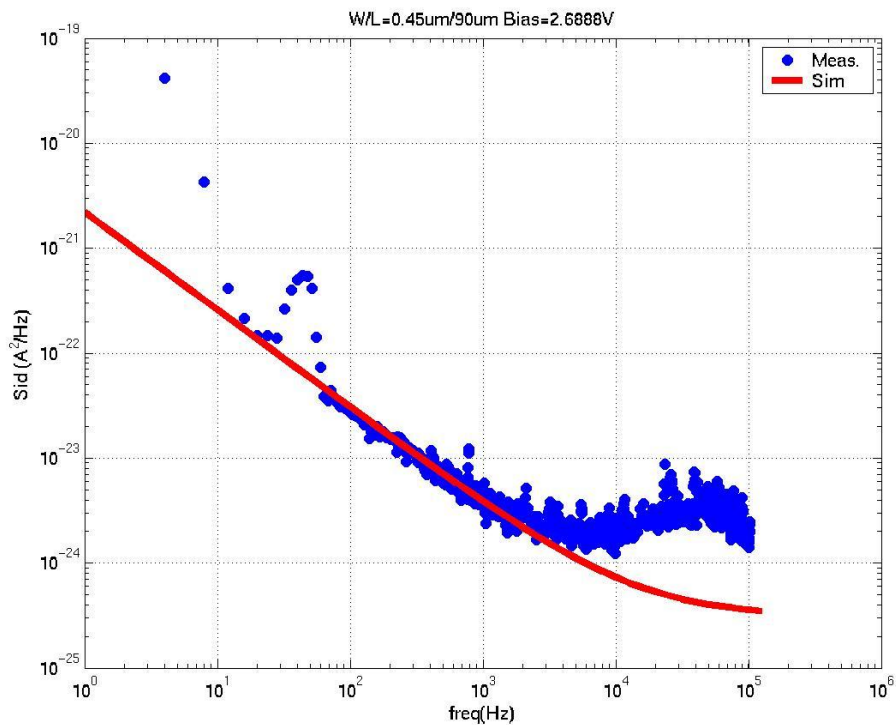


Figure 4-5 Non-Salicide P+ Poly Resistor W/L=0.45 μ m/90 μ m @ V=2.6888V

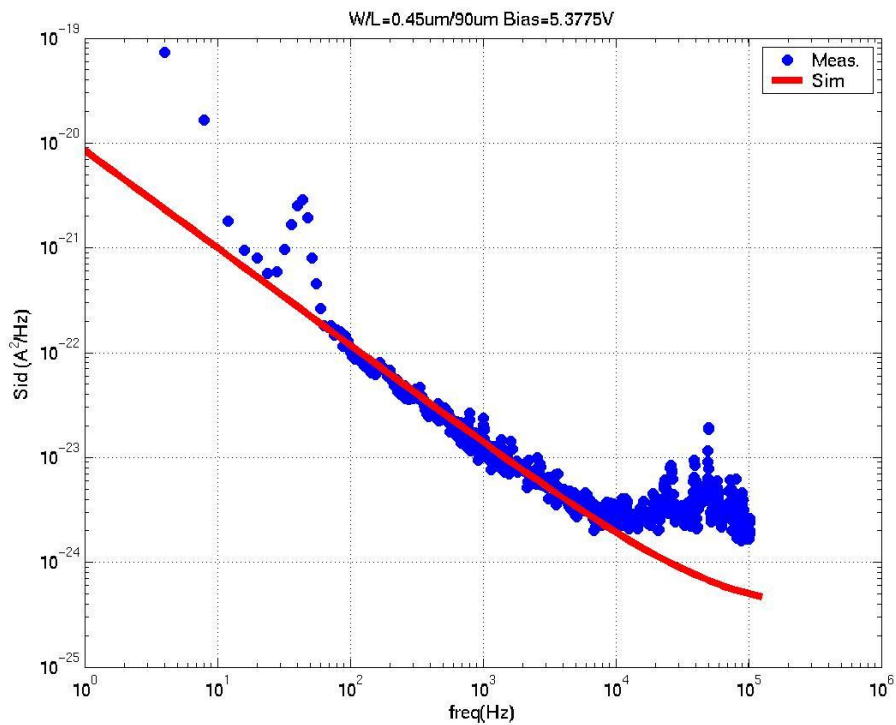


Figure 4-6 Non-Salicide P+ Poly Resistor W/L=0.45 μ m/90 μ m @ V=5.3775V

4.1.3 Non Salicide HR Poly Resistor

➤ $W=0.45\mu\text{m}$ $L=90\mu\text{m}$ (After shrink)

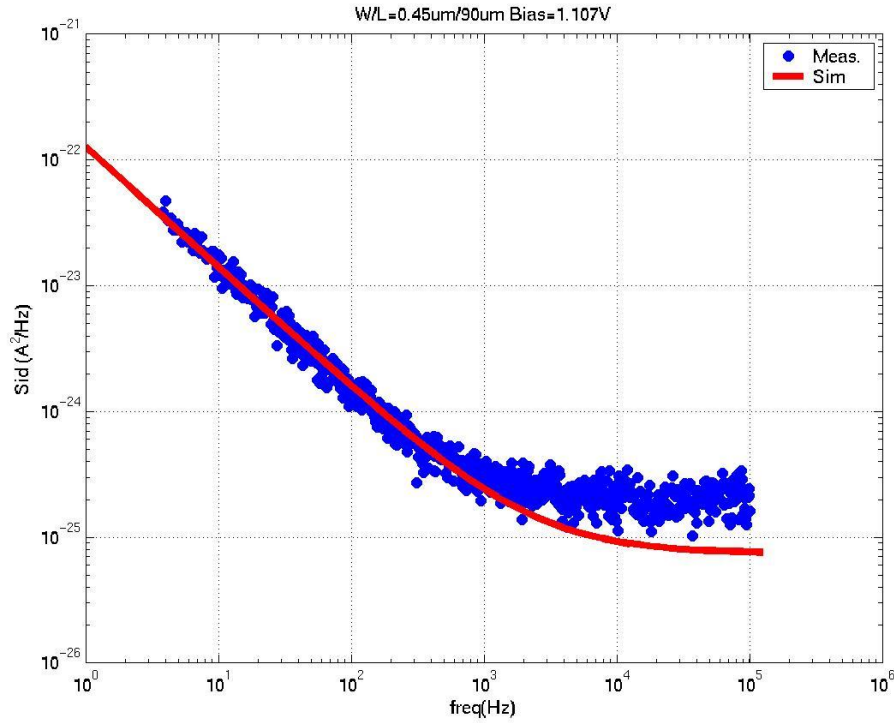


Figure 4-7 Non-Salicide HR Poly Resistor in N-WELL W/L=0.45 μm /90 μm @ V=1.107V

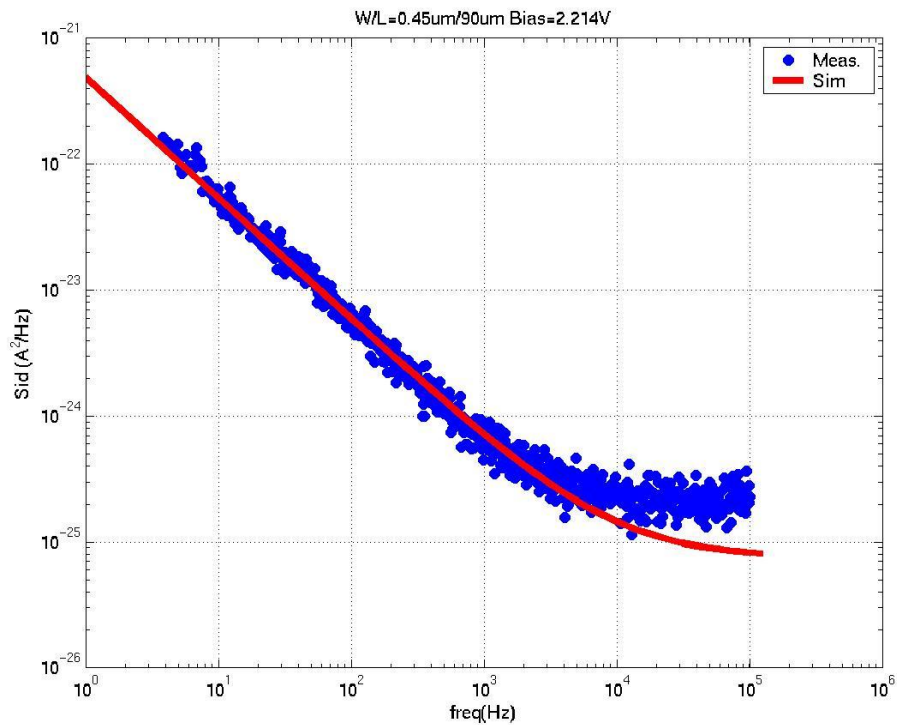


Figure 4-8 Non-Salicide HR Poly Resistor in N-WELL W/L=0.45 μ m/90 μ m @ V=2.214V

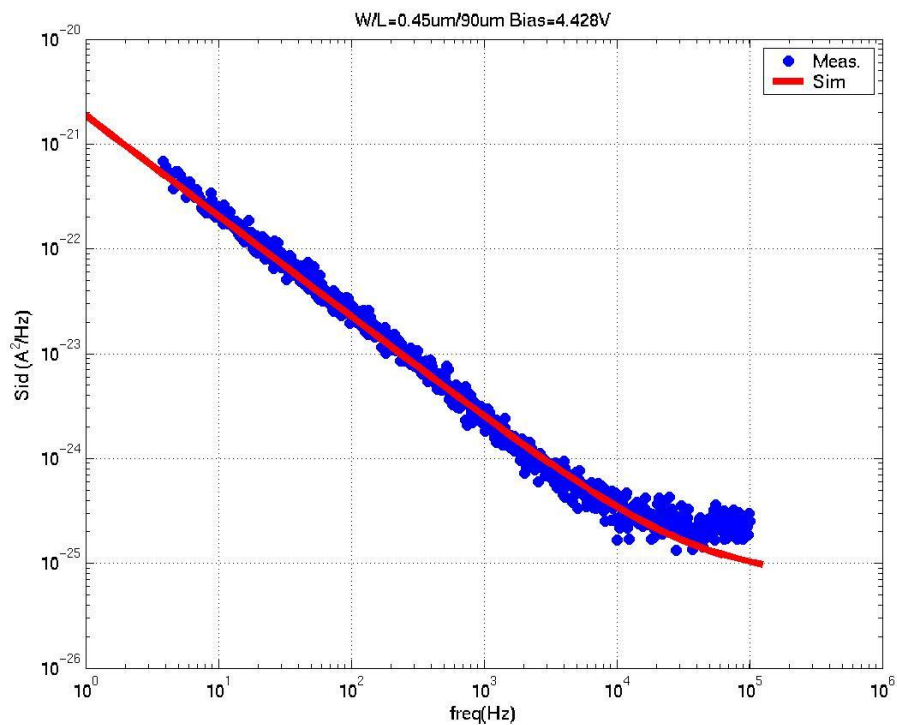


Figure 4-9 Non-Salicide HR Poly Resistor in N-WELL W/L=0.45 μ m/90 μ m @ V=4.428V

5 Appendix

5.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

Item	Warning Message	Explanation
1.	None	None

5.2 ELDO Warning Message

Table 3-2 Eldo warning message

Item	Warning Message	Explanation
1.	None	None

5.3 SPECTRE Warning Message

Table 3-3 Spectre warning message

Item	Warning Message	Explanation
1.	None	None